Matthew Lack

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Education

University of Toronto Expected May 2027
Bachelor of Science in **Computer Science** (Major) and **Mathematics** (Major) GPA: 3.7/4.0

Experience

Warp Dec 2025 | New York
Software Engineer Intern

- Designed and built end-to-end growth projects such as Warp Wrapped, a story-based year recap for all Warp users.
- Investigated and solved critical issues with legacy systems and helped migrate to Effect.ts and PostgreSQL.

Lyft May 2025 – September 2025 | Toronto, ON
Software Engineer Intern (Loyalty Team)

- Implemented new `RESTful API` endpoints in Python to support in-app gift card control and implemented matching app functionality.
- Collaborated with data engineering teams to implement event tracking for A/B experiments to measure impact on rider retention.
- Wrote and reviewed various technical specification documents to support development.
- Developed Python microservices to filter and paginate critical backend data to improve performance server-side.

MidnightSky Sep 2021 – Present | Toronto, ON
Founder, Lead Developer

- Launched a high-traffic Minecraft server with `400+` concurrent players, `10,000+` users, and `$250K+` in gross revenue.
- Led a cross-functional team of 6 developers, designers, and marketers to release major updates monthly.
- Built a distributed network of 40 game servers that function as one, using data synchronization protocols developed in-house.
- Designed robust and redundant systems to ensure users never lose progress even in the event of failures.

Qoherent.AI Sep 2024 – Dec 2024 | Toronto, ON
Software Engineer (Part-time Contract)

- Deployed `RESTful` APIs using `FastAPI`, `Python`, and `Go`, reducing data retrieval latency by 40%.
- Developed reusable and scalable UI components with `Vue.js`, `HTML`, `CSS`, and `JavaScript` to enable real-time model tracking and visualization for machine learning engineers.

Projects

Event Platform · (Kotlin, React) Jan 2026

- Created an event platform for a local nightclub that allows them to sell tickets and scan them at the door.
- Implemented role-based auth, secure mobile sms-login, timing-proof ticket scanning, comprehensive logging, and event analytics.
- Built with `Spring Boot`, `React`, `Typescript`, `Kotlin`, `JWT`, `Stripe`, and `Twilio`

iPhone X Display Reverse Engineering August 2026

- Built a 250 MHz logic analyzer on the RP2350 using eight phase-locked PIO co-processors to capture iPhone X display signals.
- Designed a custom KiCad interposer PCB and decoded MIPI DSI traffic to identify display initialization commands, brightness controls, and frame timing.
- Developed a PIO-based memory driver delivering 100 MB/s combined write bandwidth across two PSRAM chips to buffer captured signals.

FinHog · HackMIT Prize Winner (Anthropic, Warp) August 2025

- Built an AI-centric version of the popular analytics platform PostHog. FinHog analyzes your data for you and generates insightful graphs and visuals based on your data.

Skills

Java, Python, Kotlin, Javascript, TypeScript, C, C++, Go, Lua, R, Git/GitHub, Maven, Gradle, MongoDB, Postgresql, MySQL, Redis, Spring Boot, Node, Ktor, FastAPI, Django, Flask, Docker